Machine Learning Approach to Pickleball Strategy

Ryan Peters and Andy Vong

Motivation: Pickleball is an Addictive, Young Sport

Pickleball was invented
By three friends from Bainbridge Island, Washington

COVID 19 Boosted the Popularity of the Sport
This led to a massive influx of players (young and old) and investors to the sport leading to the mainstream professional play.

1965

2016

2019

2025

The Beginning of Professional Pickleball
The founding of the first professional pickleball federation

19.8 Million Players (est. 2025)
311% Growth from 2022 - 2025, 43.9% Annual Growth

Motivation: Players Inevitably Explore Pickleball Strategy

SOCIAL SPORT

Players spend hours on the courts socializing between games,not uncommon for folks to play from 6 am - 11 am every day

PLAYED IN DOUBLES FORMAT

Players must communicate and work with their partner

STRATEGY NOT SOLVED

Still a young sport and at the amateur level, players need to work with the background and limitations of their partner

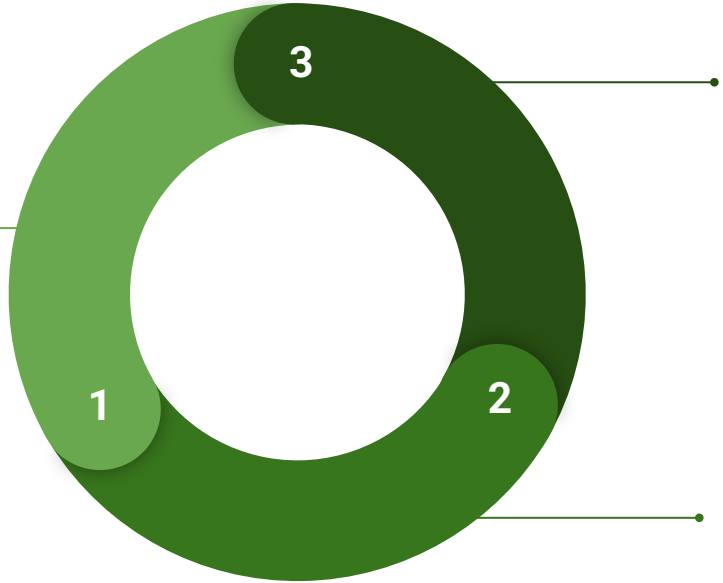
Project Goal: Inform Pickleball Strategy Using Data

Customers:
- Players
- Analysts
- Coaches

Stakeholders:
- Club owners
- Professional Organizations

Approach

Clean the Data
Ensure the data is in correct formats, handle all null values and create new features.



Train Recurrent Neural Network
Used to predict the sequence of a pickleball rally, allowing us to answer the question:
How do player choices increase or decrease the odds of winning the point?

Explore the Data
Conduct a thorough EDA to answer the following two questions:
1.Can we use all of the data?
2. Is data in format for ML modeling?

Pklmart Hosts Sufficient Dataset to Train RNN

Data comes from pklmart.
Dataset hosted on Kaggle

Contains positional (x, y coordinate) and shot type data for each shot

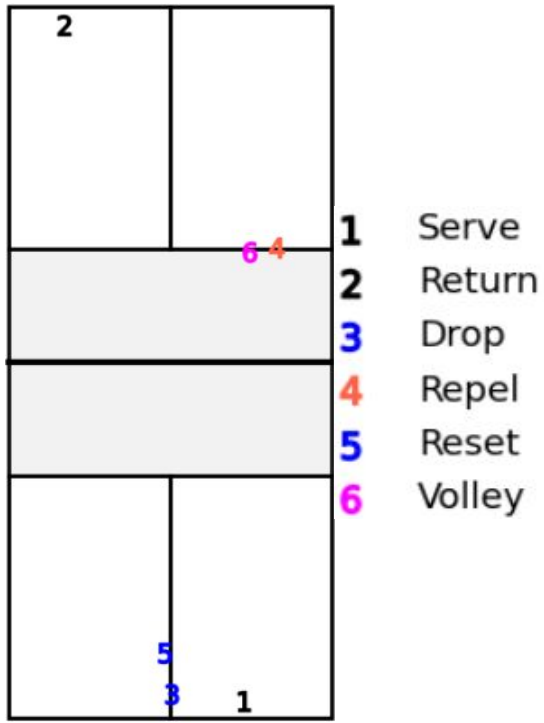
↑ **304403 SHOTS RECORDED**

↑ **39906 RALLIES**

↑ **933 GAMES**

Rationale: How We Plan to Use RNNs to Inform Strategy

- A pickleball rally can be viewed as a sequence of shots
  - [**Serve**, *Return*, **Drop**, *Repel*, **Reset**, *Volley*]
- Therefore, a predicted sequence is the same as predicting what my opponents opponents will do conditioned on what I will do
  - [**Serve**, *Return*, **Drop**, *Repel*, **Reset**, *Volley*]



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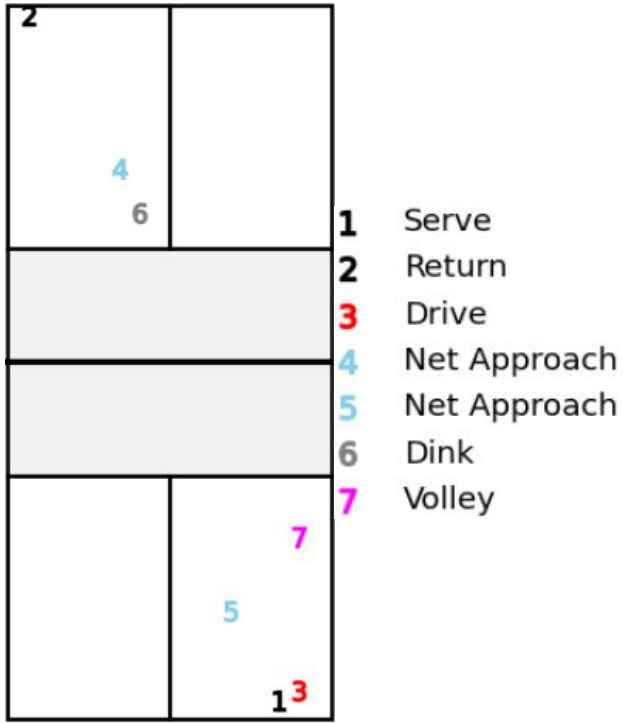
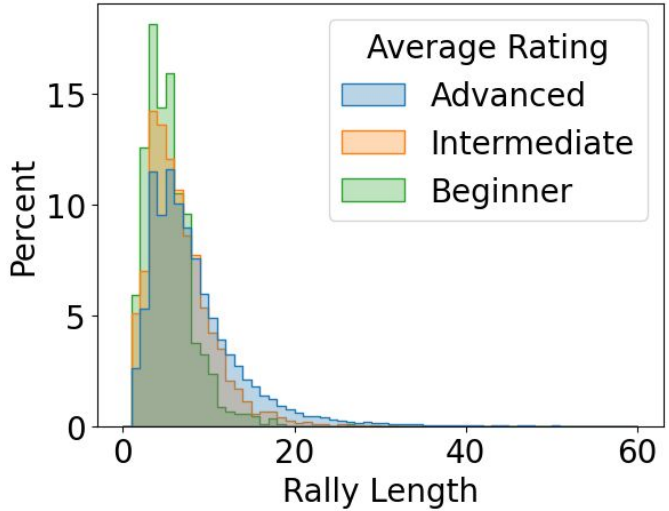
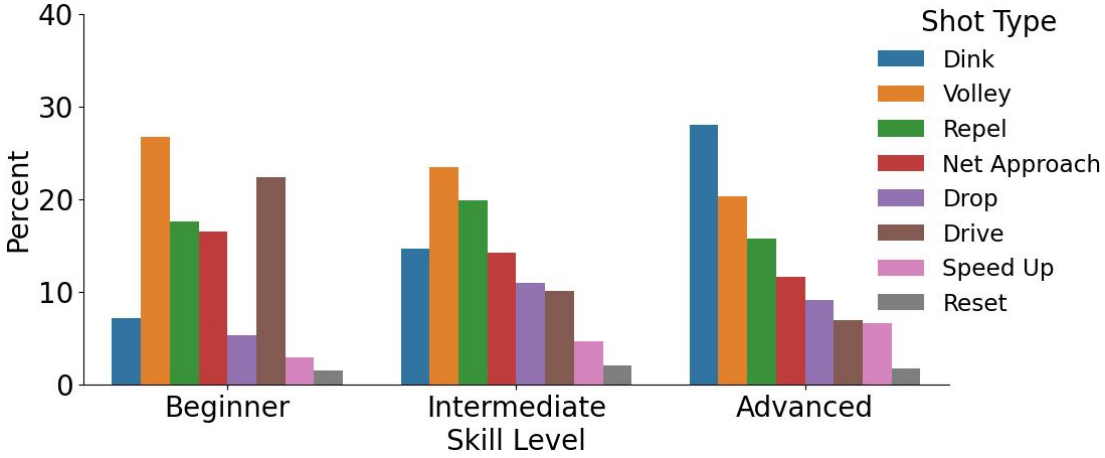
Drive, *Net Approach*, **Net Approach**, *Dink*, **Volley**]



Data Exploration: Advanced Players Are Distinguishable





- Advanced players tend to have longer rallies and play in a different style.
- We need to filter the data for the various skill levels to capture how each group plays.
- **Takeaway:** Train models specifically on skill levels to capture each level of play.

Data Exploration: Shot Type Cardinality is High

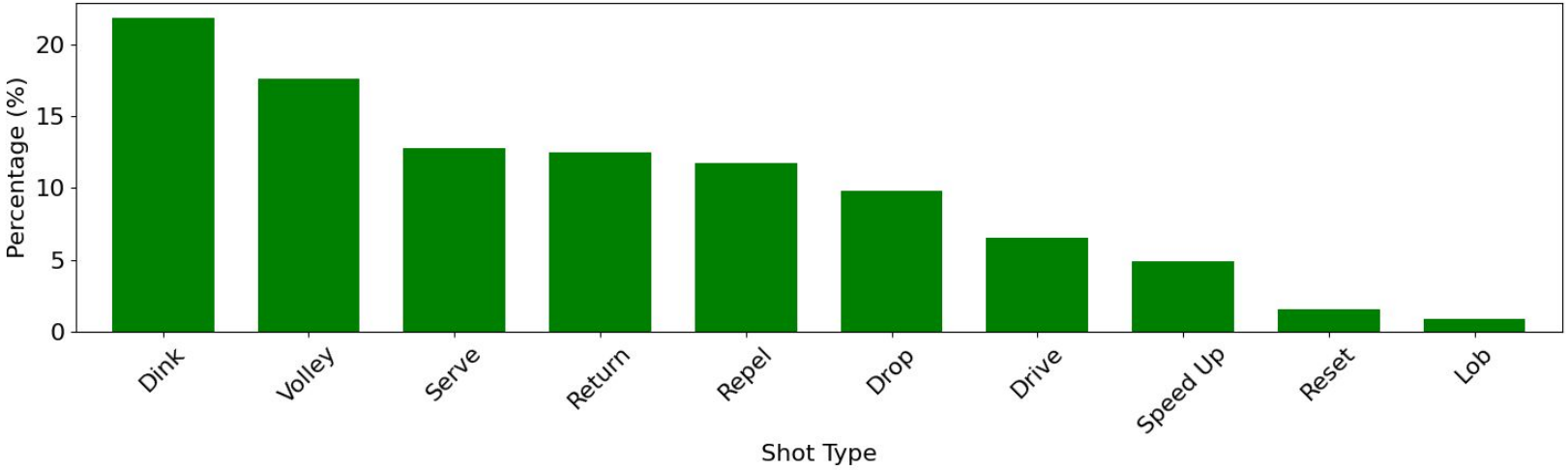
- U, O, and ball are unknown values
- Net Approach can be re-classified as other shots, especially Dink, Volley, Drop, and Drive
- **Takeaway:** Can improve chances of predicting rally sequences by re-classifying Net Approach, U, ball, and O

Feature Engineering: Training Classifier

- Model: Random forest with gradient boosting
  - Parameters:
    - n_estimators=600, max_depth=5, colsample_bytree=0.8, subsample=0.8
  - Features:
    - Positional data (x, y, Δx, Δy, shot distance, shot angle) and skill level
  - Performance:
    - **f1-score = 0.6269**

	precision	recall	f1-score	support
Dink	0.72	0.83	0.77	12015
Drive	0.51	0.46	0.48	3048
Drop	0.60	0.73	0.66	3942
Lob	0.50	0.15	0.23	499
Repel	0.66	0.73	0.69	6966
Reset	0.57	0.20	0.30	790
Speed Up	0.44	0.12	0.19	2913
Volley	0.63	0.61	0.62	9113
accuracy			0.65	39286
macro avg	0.58	0.48	0.49	39286
weighted avg	0.63	0.65	0.63	39286

Feature Engineering: Reducing Cardinality with Classifier



- **Takeaway:** Reduced cardinality down to 10 in unbiased, procedural manner
- Net approach predominantly reclassified into [Dink, Volley, Drop, Drive], matching expectation

Issues and Limitations

Lack of Shot related data

The model is constrained by a **limited feature space**. Critical physical attributes—ball speed, height, and spin—are not captured in the current dataset, preventing us from validating certain shot types.

How Shots are Classified

The dataset lacks documentation on **collection methodology**. Without knowing if shots were classified via sensors or manual entry, we cannot definitively verify the **ground truth accuracy**.

Player Movement Data

The dataset excludes **player movement metrics**, a critical component of high-level pickleball analytics. Because shot choice is heavily influenced by a player's physical position on the court

Conclusion

Project Lifecycle

1. Problem Definition
   - Project Goal: Inform Pickleball Strategy Using Data
     - Approach: Train a RNN to predict pickleball rally sequences
       - Rationale: Predicting a rally sequence is identical to predicting your opponent's decisions conditioned on your decisions
2. Data Acquisition and Understanding
   - Data Source: pklmart dataset hosted on Kaggle
   - Data Properties:
     - Advanced players are distinguishable from rest
     - Shot type contains many categories, some of which are non-sense

Conclusion

Project Lifecycle

3. Data Preparation
   - Feature Engineering:
     - (Not presented) Standardized shot positions
     - (Not presented) Created shot metrics including shot distance, shot angle, etc.
     - Reduced cardinality of shot type by reclassifying Net Approach, U, ball, and O
4. Modeling
   - Model Type: Recurrent Neural Network
     - Rationale: RNNs are suited for sequence prediction

References

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Contributions

Ryan and Andy contributed equally

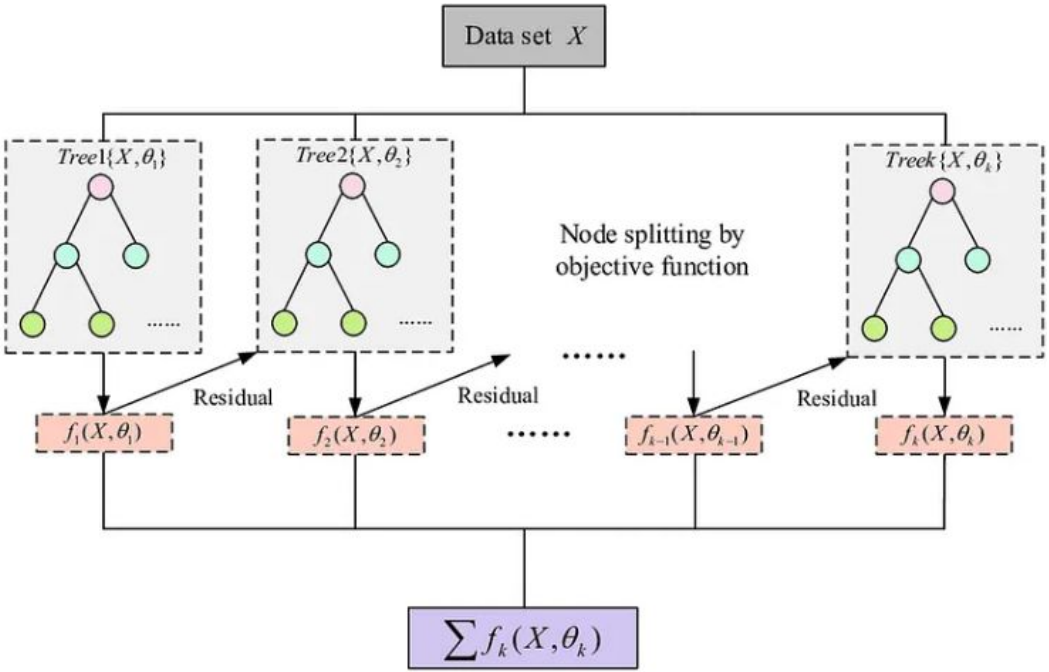
Individualized Skills:

- Ryan Peters: Experience in Applied Mathematics and working on a Data Science Team. Having been a part of the full Data Project life cycle, can assist in project organization.
- Andy Vong: Experience with writing reports and publications. Also experienced in using sklearn and TensorFlow for building and training machine learning models.

Questions?

XGBoost

- Ensemble method based on decision trees
- Residuals of prior tree are fed into the next (boosting)
  - Thus, each tree “corrects” the errors of the previous tree
- Final output is a weighted sum of each tree
  - Weights determined by learning rate and contribution to loss
- Has been highly optimized



https://medium.com/@fraidoonomarzai99/xgboost-classification-in-depth-979f11ef4bf9

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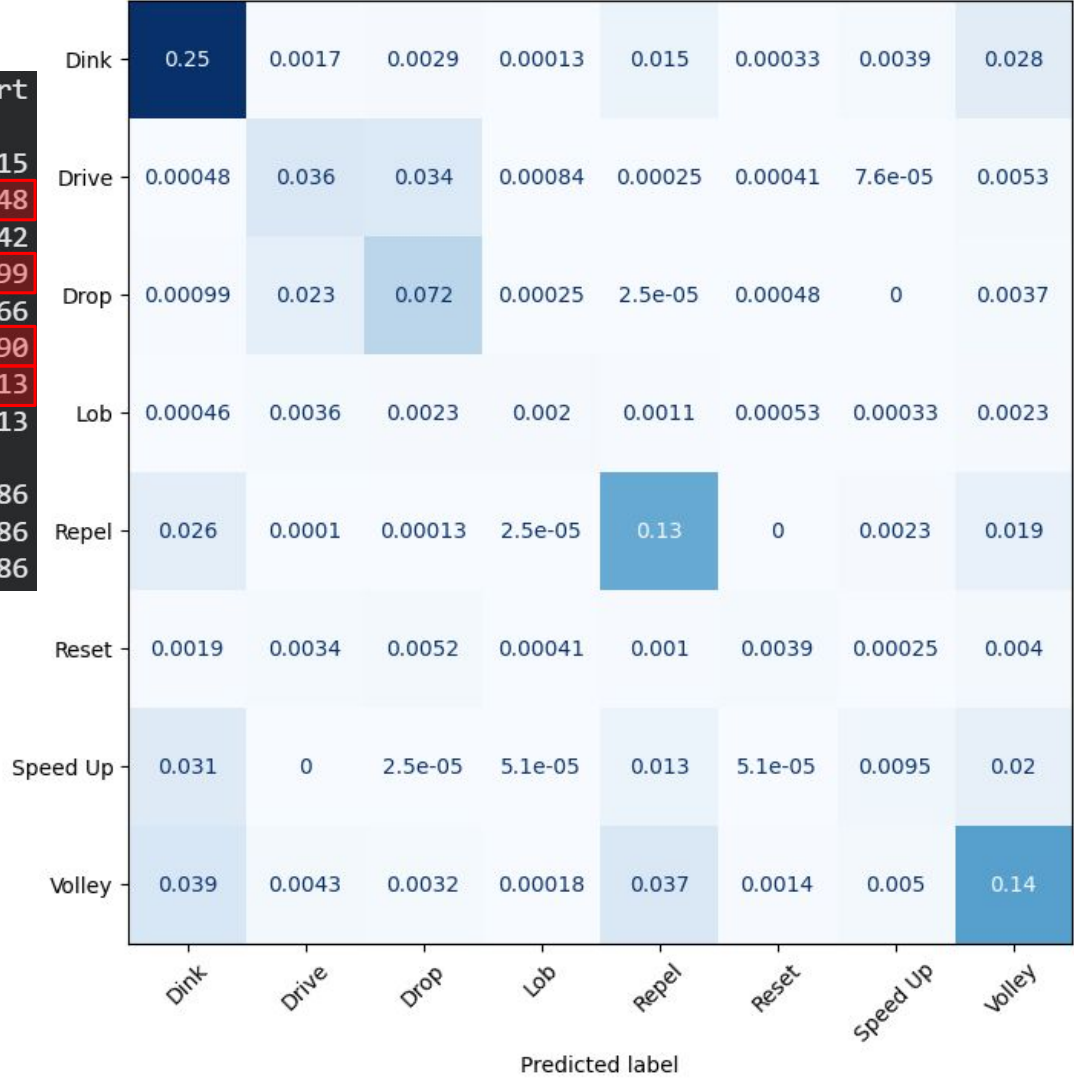
$$\text{Precision} = \frac{tp}{tp + fp}$$

$$\text{Recall} = \frac{tp}{tp + fn}$$



Approach

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  - Will enable us to ask how do player choices increase/decrease the odds of winning the point?
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  - Questions to answer:
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- Prior to training, will need to clean and explore data
  - Questions to answer:
    - Can we use all of the data?
      - No. Strategy for less experienced players will not be the same for more experienced players and data reflects the differences in players
    - Is data in format for ML modeling?
      - No. Need to reduce the number of cardinality of shot type.
        - Trained random forest classifier